C8: the D_4 symmetry cuts C_4 48×; the $k = 4$ crossover moves from 2^{150} to 2^{135}

2026-10-13. `research/gaudry_quartic_d4_20261013/`; code `gaudry_quartic_d4.rs`. Protocol registered before the measured runs. **Class: engineering.**

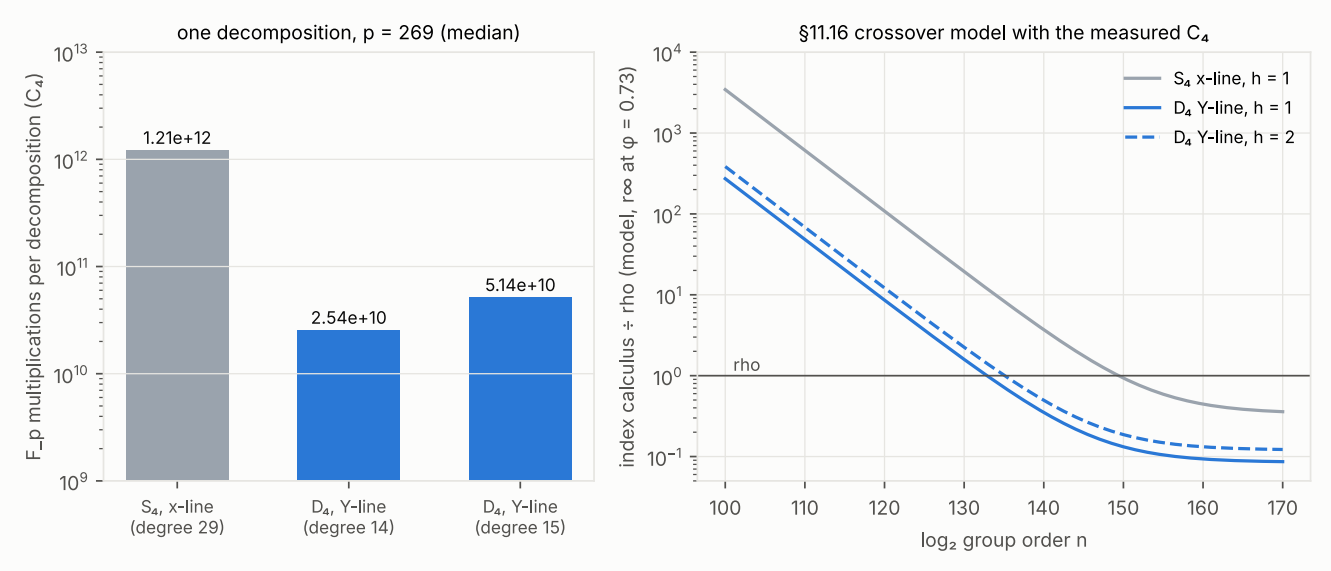
Construction

A curve over $\mathbb{F}_{p^4}$ with a rational 2-torsion point T and FGHR's coordinate Y, $Y(P+T) = -Y(P)$; base $\{P : Y(P) \in \mathbb{F}_p\}$. S_5 in Y is invariant under $D_4 = (\mathbb{Z}/2)^3 \rtimes S_4$ (order 192). In $\sigma_k = e_k(Y^2)$, $\pi = Y_1Y_2Y_3Y_4$ and $\sigma_4 = \pi^2$ the system is four quartics and $\pi^2 - \sigma_4$: Bézout 512 against 4,096, Macaulay degree 14 against 29. The invariance is checked by interpolation on fresh points.

Result, $p = 269$

	S_4, x-line	D_4, degree 14	D_4, degree 15
quotient dimension	4,096	512	512
matrix	41,780 × 40,920	13,412 × 11,628	18,527 × 15,504
C_4 ($\mathbb{F}_p$ mult.)	$1.213–1.227 \cdot 10^{12}$	$\mathbf{2.535–2.552 \cdot 10^{10}}$	$5.13 \cdot 10^{10}$
wall a solve	164–203 s	4.2–5.6 s	8.3–9.7 s

C_4 falls **47.5–48.4**×. Decompositions equal meet in the middle on all 19 solved targets; one planted target of 20 is degenerate (quotient dimension 517) at degrees 14 and 15. A uniform target decomposes on the Y-line base at 0.133 (pooled, two curves, 4,000 targets each) of the x-line rate at equal base size: the 8 even T-translates of a decomposition hit the same target.



Left: C_4 per decomposition (median). Right: §11.16's model, $S/\text{rho} = \sqrt{h}(181(p/4)C_4/(126p^2) + r_\infty/4)$, with the measured inputs.

Crossover

	n^*, $h = 1$	n^*, $h = 2$	limit past n^*
S_4, x-line	$2^{149.4}$ … $2^{152.7}$	—	0.34 … 0.63
D_4, Y-line	$2^{132.9}$ … $2^{133.3}$	$\mathbf{2^{135.1}}$ **…** $\mathbf{2^{135.8}}$	0.12 … 0.22 ($h = 2$)

Rational 2-torsion forces $h \geq 2$, so rho runs in a subgroup at least 2× smaller: the crossover moves 14–17 bits earlier. The model's exponents are extrapolated; no end-to-end S is measured. Next: the $k = 4$ ladder in ecbench.

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU, rustc 1.97.0. Conductor unreachable: no task card.